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Jiayu WANG

PhD in mechanical engineering

EXPERIENCE

- 12.2025 **Laboratoire Interdisciplinaire de Physique, UGA** *Saint-Martin-d'Hères, France*
Postdoctoral researcher
Supervisor: E. Siéfert
Subject: Dynamic control of Gaussian morphing structures.
Theoretical and experimental study.
- 10.2022 - 11.2025 **L'institut Jean Le Rond d'Alembert, Sorbonne Université** *Paris, France*
PhD student
Supervisor: S. Neukirch and A. Antkowiak
Subject: Buckling and packing of beams and plates inside liquid cavities.
Theoretical and numerical study.
- 04.2022 - 09.2022 **Process & Energy, 3ME Faculty, TU Delft** *Delft, The Netherlands*
Research Internship / Final year project
Supervisor: B. Bugeat and R. Pecnik
Subject: Transition to turbulence in supercritical fluids
Simulation and analysis of secondary instabilities for boundary layer flow of a supercritical fluid using DNS
- 05.2021 - 08.2021 **Institut Jean Le Rond D'Alembert, Sorbonne Université** *Paris, France*
Research Internship
Supervisor: A. Antkowiak
Subject: Study of hydrothermal vents' formation and interaction
Theoretical and experimental study
- 11.2019 - 12.2019 **ISIR, Sorbonne Université** *Paris, France*
Research Internship
Supervisor: L. Saint-Bauzel
Subject: Bio-mechanical assessments of an arm exoskeleton
Using software *OpenSim*, simulating and analysing mechanical effects
- 05.2018 - 06.2018 **Hangzhou Tiankuan Technology** *Zhejiang, China*
R&D department Assistant
Supervisor: Ming Geng
Subject: Professional discovery internship.
Observation of the work in R&D department. Learning about the product management, patent application, etc.

EDUCATION

- 2022 - 2025 **Doctorate: Mechanical Engineering** *Paris, France*
Sorbonne Université
Buckling and packing of beams and plates inside liquid cavities

2020 - 2022	Master's Degree: Computational Mechanics <i>Sorbonne Université</i> The program covers fundamental and advanced topics on solid and structural mechanics, fluid mechanics, and their interactions.	<i>Paris, France</i>
01.2020 - 05.2020	Exchange semester in NUS <i>National University of Singapore</i> Courses: mechanics of materials, intermediate fluid mechanics, sustainable energy conversion, computational methods in physics, probability and statistics	<i>Singapore</i>
2017 - 2020	Bachelor's degree: Mechanical Engineering <i>Sorbonne Université</i> Cursus Master en Ingenierie (CMI) Special program CMI: 5 years intensive and selective education program, an international benchmark of master's of Science in engineering.	<i>Paris, France</i>

SCIENTIFIC PARTICIPATIONS

07.2024	Journées d'Alembertiennes Organizing committee	<i>Saint-Cyr, France</i>
06.2024	Fluid and Elasticity conference Contributed talk	<i>Arcachon, France</i>
05.2024	The European Mechanics of Materials Conferences (EMMC) Contributed talk	<i>Madrid, Spain</i>
03.2024	Rencontre du non-linéaires (RNL) Contributed talk	<i>Paris, France</i>
10.2023	MePhy Day: Surface tension, geometry and soft solids Contributed talk	<i>Paris, France</i>
06.2023	CISM Summer course (Interfacial Flows: The Power and Beauty of Asymptotic Methods) Poster	<i>Udine, Italy</i>

TEACHING

2022 - 2025	Sorbonne Université <i>Chargée de Mission d'Enseignement (64h per year)</i> LU1MEPY2 : Mécanique - Physique 2, Travaux Pratiques. (2022/23) LU2ME002 : Bases de la thermodynamique et thermique, Travaux Pratiques. (2022/23, 2023/24) LU3ME004 : Bases de la mécanique des milieux continus, Travaux Pratiques. (2022/23, 2023/24) LU2ME006 : Analyse vectorielle et intégrales multiples, Travaux Dirigés. (2023/24, 2024/25)	<i>Paris, France</i>
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TECHNICAL STRENGTHS

Computer Languages	Python, C, MATLAB, Mathematica
Numerical Simulation	OpenFoam, FEniCS, CATIA, SolidWorks, Ansys Fluent
Tools	CasADi, L ^A T _E X, git

LANGUAGES

Chinese : Native	French : Fluent	English : Fluent
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